TASK 2 – Website Security Assessment for Client Category: Cybersecurity

🔹 Part A: Client Understanding & Planning

1️⃣ Client Scenario Report

Client Scenario Report – CoreTech Innovations

Website security is a critical requirement for modern businesses because websites store sensitive information such as customer data, login credentials, and payment details. Any security breach can lead to data loss, financial damage, legal issues, and loss of customer trust. For business clients, even a small vulnerability can be exploited by attackers, causing serious reputational harm.

Common website threats include **SQL Injection**, where attackers manipulate database queries to access or modify sensitive data; **Cross-Site Scripting (XSS)**, which allows malicious scripts to run in users' browsers; and **Brute Force Attacks**, where attackers repeatedly attempt login credentials to gain unauthorized access. Other risks include outdated plugins, weak passwords, and misconfigured servers, which increase the attack surface of a website.

Cybersecurity services help protect business clients by identifying vulnerabilities before attackers can exploit them. Regular security scans, vulnerability assessments, and security audits help ensure that websites follow best security practices. These services also assist in implementing secure authentication, encryption, firewall protection, and timely patch management.

CoreTech Innovations can offer basic website security services such as vulnerability scanning, open port analysis, HTTPS verification, and security configuration reviews. CoreTech can also provide clients with clear security reports, risk explanations, and preventive recommendations. By offering proactive website security assessments, CoreTech helps business clients maintain secure, reliable, and trustworthy online platforms.

🔹 Part B: Practical / Client Security Assessment

🎯 Practical Task

Perform a basic security assessment on a test website (use demo website, localhost, or legally allowed target only).

🔐 **Website Security Audit Report**

Client: CoreTech Innovations
Assessment Type: Manual Website Security Audit
Target Website: Demo / Test Website (Localhost or Legally Allowed Website)
Audit Level: Beginner (Client Awareness & Basic Security Review)

🔹 **STEP 1: Manual Security Audit**

This manual security audit was conducted to identify common security weaknesses that may exist on a typical business website. No automated exploitation or illegal testing was performed. The assessment focuses on configuration-based and visible security issues that are commonly found in small and medium-scale websites.

The audit specifically examined password security practices, HTTPS implementation, authentication mechanisms, and software/plugin update status. These areas are frequently targeted by attackers because they are often misconfigured or neglected.

All testing was carried out in a controlled environment using a demo website or localhost setup. The purpose of this audit is to highlight potential risks and help CoreTech Innovations understand how basic cybersecurity checks can protect business clients from common cyber threats.

🔹 STEP 2: List of Vulnerabilities Found

Below is the **manual vulnerability list** identified during the website security audit. These are common security issues observed on many business websites.

🔴 1. Weak Passwords

- Simple and predictable passwords used for admin or user accounts
- No password complexity policy enforced
- No account lockout after multiple failed login attempts

Impact:
Attackers can use brute-force or credential-stuffing attacks to gain unauthorized access.

🔴 2. Missing HTTPS

- Website is accessible over HTTP
- No SSL/TLS certificate installed
- Data transmitted in plain text

Impact:
Sensitive information such as usernames and passwords can be intercepted by attackers using man-in-the-middle attacks.

🔴 3. Poor Authentication Mechanisms

- No two-factor authentication (2FA)
- Session timeout not configured properly
- Login error messages reveal whether the username exists

Impact:
Weak authentication increases the risk of account compromise and session hijacking.

🔴 4. Outdated Plugins / Software

- CMS plugins not updated to latest versions
- Unused plugins still enabled
- Server software running older versions

Impact:
Outdated components may contain known vulnerabilities that attackers can exploit easily.

🔹 STEP 3: Prevention Recommendations

The following recommendations are provided to reduce security risks and improve the overall security posture of the website.

✅ 1. Strengthen Password Security

- Enforce strong password policies (minimum length, complexity)
- Enable account lockout after multiple failed login attempts
- Encourage the use of password managers
- Regularly change administrative passwords

✅ 2. Enable HTTPS Encryption

- Install a valid SSL/TLS certificate
- Redirect all HTTP traffic to HTTPS
- Enable HTTP Strict Transport Security (HSTS)

✅ 3. Improve Authentication Mechanisms

- Implement Two-Factor Authentication (2FA)
- Configure proper session timeout and automatic logout
- Use generic login error messages to avoid information disclosure

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✅ 4. Maintain Updates & Patching

- Regularly update CMS, plugins, and themes

- Remove unused or unnecessary plugins
- Monitor software vendor security updates

🔹 STEP 4: Short Summary

This manual website security audit identified several common security weaknesses, including weak password practices, missing HTTPS encryption, poor authentication mechanisms, and outdated plugins or software components. These vulnerabilities can expose business websites to attacks such as brute-force attempts, data interception, and unauthorized access.

Although no active exploitation was performed, the presence of these issues indicates a moderate security risk if left unaddressed. Implementing strong password policies, enabling HTTPS, improving authentication controls, and maintaining regular software updates will significantly enhance website security.

CoreTech Innovations can use this audit approach to provide basic security assessments for business clients and help them improve their overall cybersecurity posture.